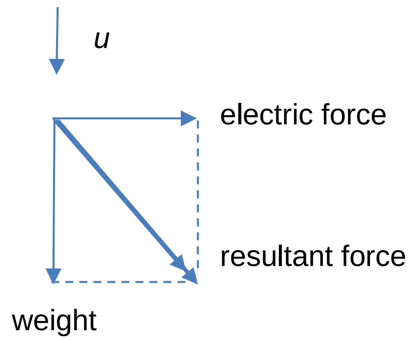
5	
(a)	Electric force $= q\dfrac{\Delta V}{d}$ $= 0.78 \times 10^{-6}\,\dfrac{2000}{5.0}$ $= 3.12 \times 10^{-4}$ N
	Direction of electric force is rightwards towards Q.

(b)(i)	 Forces must be labeled clearly.
(b)(ii)	The weight of particle X has the same order of magnitude as the electric force.
(c)	Horizontally, $s_x = 5.0, u_x = 0, a_x = \dfrac{3.12 \times 10^{-4}}{\frac{4.9 \times 10^{-4}}{9.81}} = 6.25 \text{ m s}^{-2}$
	Using $v_x^2 = u_x^2 + 2a_x s_x$, $v_x = 7.91 \text{ m s}^{-1}$
	Using $s_x = \dfrac{1}{2} a_x t^2$, $t = 1.26 \text{ s}$ Vertically, Using $v_Y = u + a_y t$ $v_Y = 23 + (9.81)(1.26)$ $= 35.4 \text{ m s}^{-1}$
	Magnitude of resultant velocity $= \sqrt{7.91^2 + 35.4^2}$ $= 36.3 \text{ m s}^{-1}$ (3 s.f.)
(d)	Since electric force and weight are of constant magnitudes and directions, the resultant force would also be of constant magnitude and direction.
	As the velocity has a component along the resultant force and a component perpendicular to it, the path taken by the particle would be parabolic.
(e)	$\Delta V = 2000 \text{ V}$
	Change in EPE $= q\Delta V$ $= (-0.78 \times 10^{-6})(1000 - (-1000))$ $= -1.56 \times 10^{-3} \text{ J}$
6	